



# Capstone Architecture + AI Spec v1

Week 8 • Day 3 • AI Spec Development & Capstone

TPM Academy



## Day 3 Objectives

- Produce **Technical Concept Document (TCD-light)** in 40 minutes (Week-4 compressed)
- Produce **Technical Modeling Document (TMD-light)** in 40 minutes (Week-5 compressed)
- Run the **5-prompt AI Spec sequence** on the capstone, drawing on Day-2's Product Requirements Document (PRD-light)
- Cross-quad **spot-check** the AI Spec against sources
- Plan Day-4 validation work



# Today's Shape

| Clock         | Block                       | Outcome                             |
|---------------|-----------------------------|-------------------------------------|
| 09:00 – 09:15 | Opening                     | Pin PRD-light + Day-3 plan          |
| 09:15 – 10:30 | Block 1 + Activity 1        | TCD-light                           |
| 10:45 – 12:00 | Block 2 + Activity 2        | TMD-light                           |
| 13:00 – 14:15 | Block 3 + Activity 3        | 5-prompt AI Spec sequence           |
| 14:30 – 16:00 | Block 4 + Activity 4 + Wrap | Cross-quad spot-check + Day-4 setup |

# Block 1 — TCD-Light

Week 4 in 45 minutes



# TCD-Light (1.5 pages)

```
## 1. Architecture stance – 1 paragraph  
(deploy / failure / scale)  
## 2. Integration map – table (system / owner / sync-async /  
R-W / failure handling)  
## 3. Threat-model summary – 3 STRIDE threats + mitigations  
## 4. SLOs – 1 latency / 1 availability / 1 rate-limit  
(target + defense + verification)  
## 5. Top 3 trade-offs – Option A/B/Choice/Cost/Revisit  
## 6. Stakeholder sign-off – 3-5 stakeholders + status
```

# Activity 1 — TCD-Light

Quads • 45 minutes

- 10 min: architecture stance (3-question frame)
- 10 min: integration map
- 10 min: the STRIDE model (Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege) summary (3 threats)
- 5 min: 3 Service Level Objectives (SLOs)
- 5 min: top trade-offs



10 + 10 + 10 + 5 + 5

# Block 2 — TMD-Light

Week 5 in 50 minutes



# TMD-Light (1.5 pages)

```
## 1. Data model – 3-5 entities (PK + indexes + relationships)
+ 1 storage trade-off
## 2. Cloud topology – region / managed-vs-self / tenancy /
network / ROM cost
## 3. API contract – 3-5 endpoints (method + path + status +
idempotency)
## 4. Sequence – happy + 1 sad + 1 weird with named invariant
## 5. Performance baseline + monitoring – 3 baselines + 3 alerts
+ 1 leading indicator
```



# Activity 2 — TMD-Light

Quads • 50 minutes

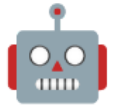
- 10 min: data model + 1 trade-off
- 10 min: cloud topology + rough order of magnitude (ROM) cost
- 10 min: API contract
- 5 min: sequence (happy + sad + weird with invariant)
- 5 min: baselines + alerts + leading indicator



10 + 10 + 10 + 5 + 5

# Block 3 — 5-Prompt AI Spec Sequence

Apply Day-1's pattern to the capstone



# The 5 Prompts

| # | Job                      | Inputs                                                           |
|---|--------------------------|------------------------------------------------------------------|
| 1 | Headline + summary       | PRD-light Sections 1–5, TCD-light Section 1, TMD-light Section 3 |
| 2 | Data + API synthesis     | TMD-light Section 1 + Section 3                                  |
| 3 | Sequence + failure       | TMD-light Section 4                                              |
| 4 | Constraints              | TCD-light Section 3 + Section 4 + PRD-light Section 7            |
| 5 | Decisions + stakeholders | TCD-light Section 5 + Section 6                                  |

**Validate between every prompt.** The chain is what keeps the AI Spec honest.

## Validation Discipline (under time pressure)

- Read every output against the source
- Mark: cross-checked / spot-checked / pending / rejected
- Cut hallucinated claims
- **Preserve verbatim:** trade-off costs, revisit triggers, named invariants, customer quotes

**Time pressure tempts skipping validation.** Hold the line. Catch this in the room.

# Activity 3 — AI Spec v1

Quads • 50 minutes

- 8 min × 5 prompts = 40 min total
- Per prompt: paste inputs → capture output → validate → mark



8 × 5 prompts

# Block 4 — Cross-Quad Spot-Check + Day-4 Setup

Catch what the author quad missed

# Cross-Quad Validation Protocol

1. Pair quads
2. Reviewer quad receives **source artifacts + AI Spec v1**
3. 20 min validation: **5–10 specific claims** from AI Spec checked against source
4. Mark: **wrong / partly right / right**
5. Author quad listens; revises

**Watch for rubber-stamping.** The reviewer should find at least 1 weak claim.

# Activity 4 — Cross-Quad + Day-4 Plan

Quad pairs • 55 minutes • Wrap

- 5 min: pair set-up
- 20 min: validation pass on 5–10 specific claims
- 5 min: author listens
- 15 min: author revises



5 + 20 + 5 + 15 • 15-min wrap





# Day 3 Wrap

## What we built

- TCD-light (1.5 pages)
- TMD-light (1.5 pages)
- AI Spec v1 (assembled from 5 prompts)
- Cross-quad validation findings

## What opens tomorrow

- **Technical logic validation + finalization**
- Catch inconsistencies across the artifact set
- Polish for Friday
- Rehearse the presentation

**Bring to Day 4:** All 4 documents + cross-quad findings. Tomorrow we make them ship-ready.

# Questions & Reflection

Where in your AI Spec do you suspect there's a hallucination you didn't catch?